

Week 2

COGS 108 Wi 2025

Due dates

- D2: Due on Sunday (Jan 26th)
- Q3: Due on Sunday (Jan 26th)
- PLEASE ACCEPT YOUR GITHUB INVITATION
 - Check your mailbox, accept invitation
 - This is necessary for your group project

Datahub/JupyterLab

Do not forget to submit(time-stamps)



Submitted assignments		
Practice_COGS108_FA24	COGS108_FA24_A00	Fetch Feedback
2024-09-28 07:25:28.062723 UTC		
2024-10-02 23:16:02.164214 UTC		
2024-10-04 05:57:15.640169 UTC		
PythonReview_COGS108_FA24	COGS108_FA24_A00	Fetch Feedback
2024-10-01 02:44:25.028562 UTC		

Important things to note

- Validate != Submit
- might need to optimize your code if runs too long
- Do not re-submit after due dates
- Do not delete the original cell

Datahub/JupyterLab



Do not re-submit after due dates

- Please double check before submitting
- JupyterLab won't store your code

Do not delete the original cell

- autograde based on cell ID

Note: Make sure you provide your GitHub username, **not the email address** you used to create the account, follow instructions, and avoid typos. You will not get credit for this section if typos are made.

In [1]: ID: cell-784114344a572182 Autograded answer

```
### BEGIN SOLUTION
PID = 'A1234567'
github_username = 'ShanEllis'
### END SOLUTION
```

In [2]: -

```
# if you navigate to this link
# it should open your GitHub page
# if it doesn't, fix your username above
'www.github.com/' + github_username
```

Out[2]: 'www.github.com/ShanEllis'

In [3]: Points: 0 ID: cell-216ded1c5ab2c280 Autograder tests

```
assert PID
assert github_username
```

ID: part2-intr Read-only

Part 2: Python (5.5 points)

GitHub



1. Fork

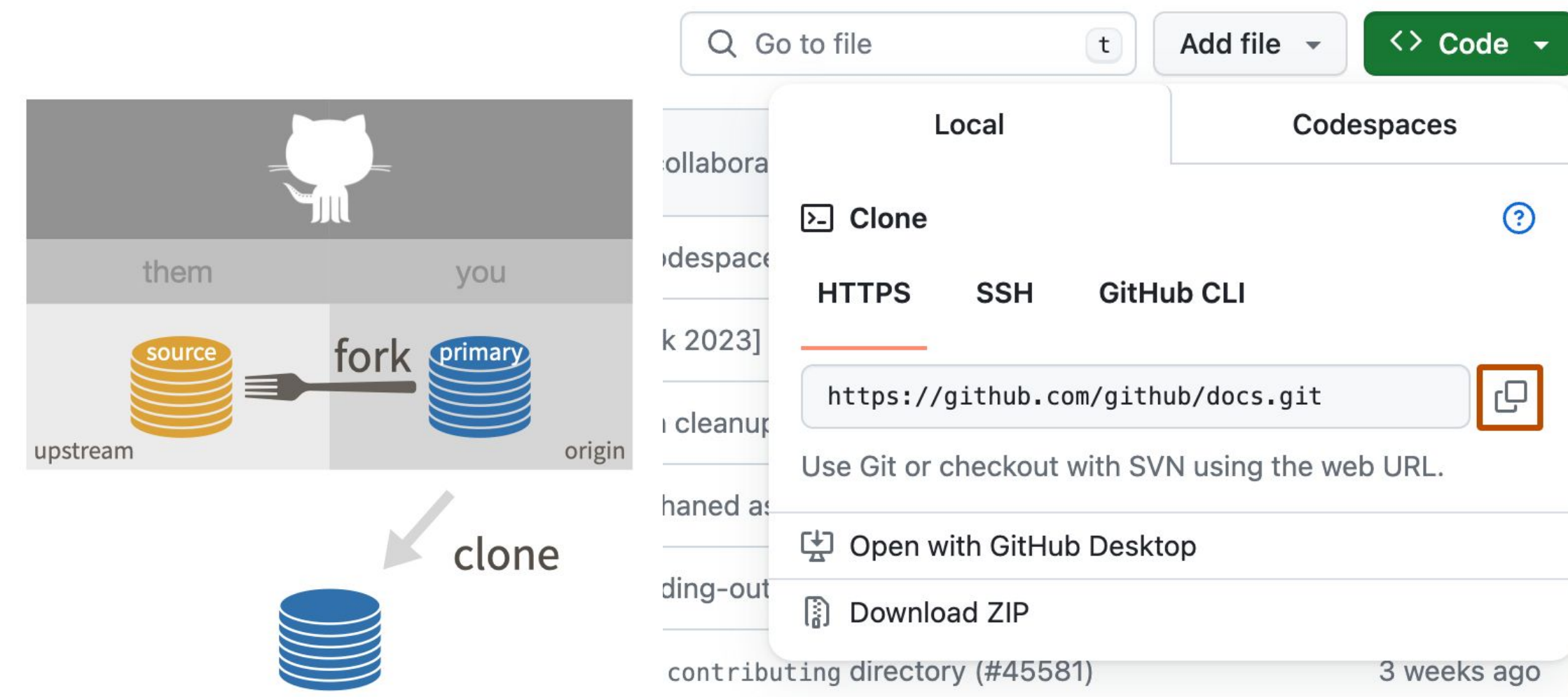
A “fork” is just an independent copy of a repository that you can develop on without affecting the original.

2. Clone

Creates a local copy of a repository on your machine.

3. You can “push” changes back to the remote repository if you have permissions.

```
$ git clone https://github.com/user/repo.git
```

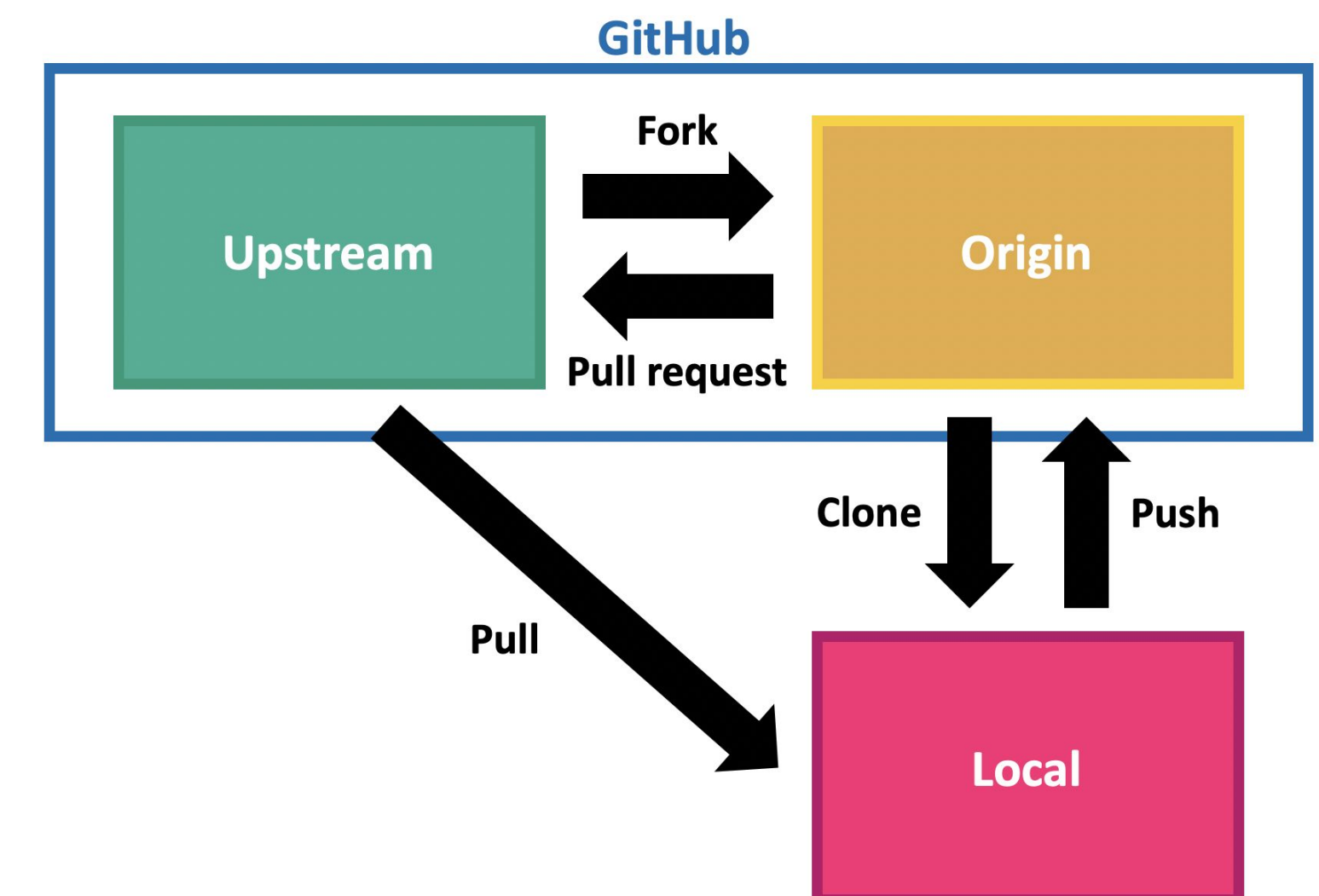
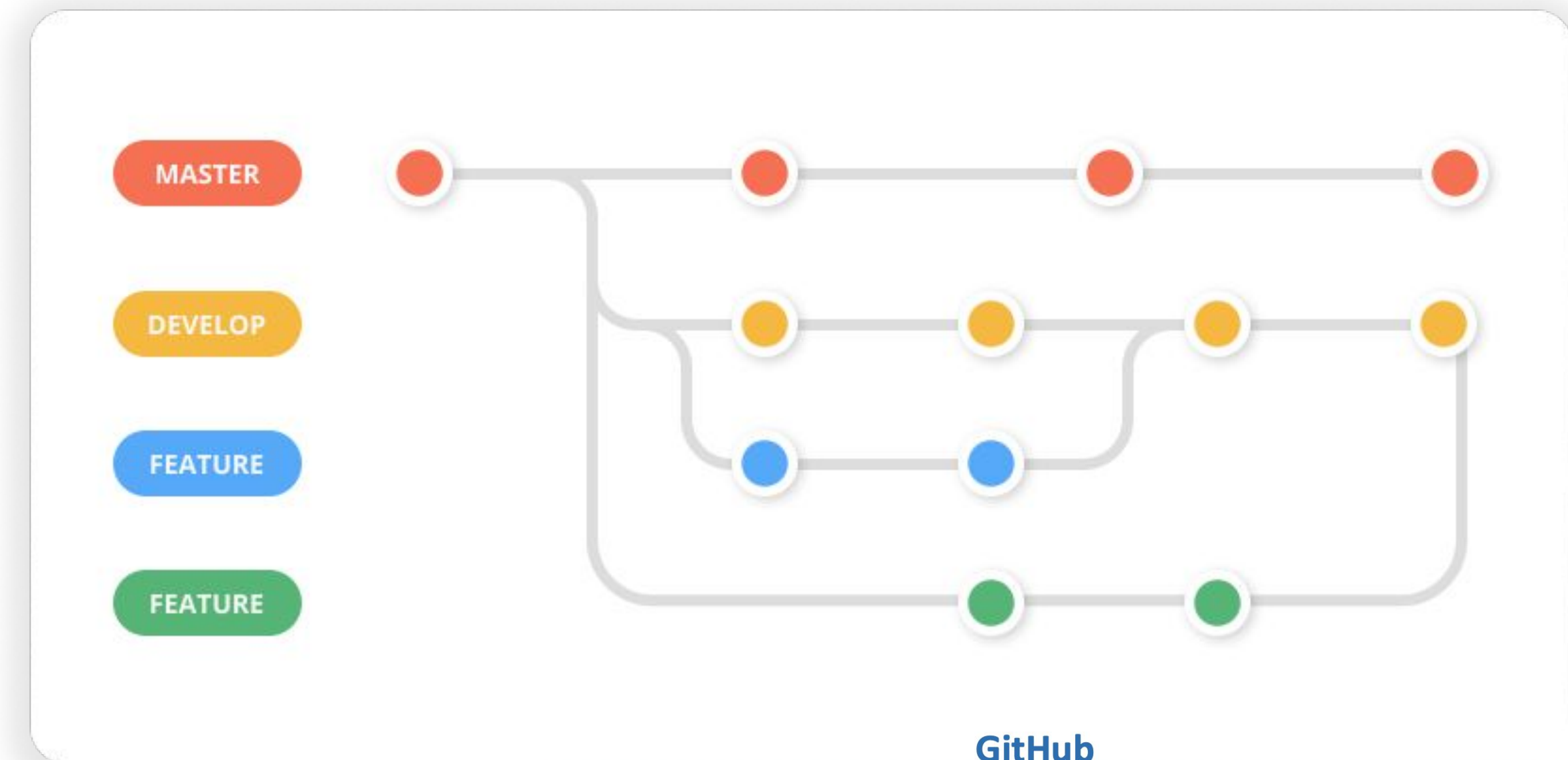


Branching

Branch will be independent of the clean, functioning “master” code and is a safe place for you to delete, modify and add code.

Pull Request

Asks the maintainers of the original repository to take a look at and hopefully integrate your code changes into their repository.



GitHub Issues are used to track tasks, enhancements, and bugs for a project, allowing team members to discuss and manage work collaboratively.

GitHub Projects is a tool to organize and track the progress of issues and pull requests in a board view, helping teams visualize workflows and prioritize tasks.

🔒 OctoArcade Invaders

📅 Planning

📊 Sprint Board

📋 Alpha

📅 Roadmap

📅 My work

📅 Features

📅 Priority

📅 By person

📊 Status Board

📊 By status

📅 By Sprint

📅 Done

Filter by keyword or by field

🕒 Not Started19Estimate: 37

🕒 planning-tracking-demo #810

Beta go-no-go meeting

🕒 planning-tracking-demo #800

Save score across levels

🕒 planning-tracking-demo #784

Interviews with media outlets

epic

🕒 Draft

Enable for teams

🕒 planning-tracking-demo #1161

tweak difficulty

🕒 planning-tracking-demo #1167

Update README.md

🕒 Draft

Prevent the Konami code from bringing down all of GitHub

+ Add item

📅 Planning19Estimate: 109

📅 planning-tracking-demo #823

Updates and bug fixes to engine from Beta

bugdemo

📅 planning-tracking-demo #824

Beta signup page

need help

📅 planning-tracking-demo #806

[Tracking] Upsell / Growth experience

backlogfeature

📅 planning-tracking-demo #818

Account subscription design

📅 planning-tracking-demo #828

Acquire domain for launch

📅 planning-tracking-demo #832

Final creative shots from game

📅 planning-tracking-demo #829

+ Add item

🔨 Building8Estimate: 40

🔨 planning-tracking-demo #1160

Update documentation

🔨 planning-tracking-demo #814

Updates to collision logic

enhancement

🔨 planning-tracking-demo #816

Free and paid levels

need help

🔨 planning-tracking-demo #831

Documentation and Support

need help

🔨 planning-tracking-demo #821

Updates to alien, beam, bomb and cannon sprites

#370

🔨 planning-tracking-demo #802

Updates to velocity of the ship and alien movements

+ Add item

🏁 Review5Estimate: 17

🏁 planning-tracking-demo #822

Hero site - Development

#12#1160in-reviewtaskurgent

🏁 planning-tracking-demo #808

General bug fixes from Alpha feedback

#992

🏁 planning-tracking-demo #1151

Design new launch screen

#374web

🏁 planning-tracking-demo #793

Polished alien, beam, and cannon sprite files

🏁 planning-tracking-demo #1101

[Tracking] Integrate payments system

backlogfeature

+ Add item

🔒 OctoArcade Invaders

📅 Planning

📊 Sprint Board

📋 Alpha

📅 Roadmap

📅 My work

📅 Features

📅 Priority

📅 By person

📊 Status Board

📊 By status

📅 By Sprint

	Title	Status	Assignees	Priority	Labels
▼	Squad 16				
1	🕒 [Tracking] Allow users to perform a manual resetting #1100	Done✔️	shiftkey	★★★★ High	duplicate enhancement epic f
2	🕒 Integrate with Commerce Service #3	Done✔️	rileybroughten	★★★★ High	
3	🕒 Hero site - Development #822	Review🔴	azenMatt	! Urgent	in-review task urgent
4	🕒 Integrate with backend systems #809	Done✔️	keisaacson	! Urgent	demo prototype urgent
5	🕒 Prevent the Konami code from bringing down all of GitHub	Not Started🕒			
6	🕒 investigate latency	Not Started🕒			
+ Add item					
▼	Squad 29				
7	🕒 Integrate with Leaderboard Service #811	Done✔️	dusave and jclem	★★★★ High	enhancement
8	🕒 [Tracking] Upsell / Growth experience #806	Planning📅	mariorod	★★★ Medium	backlog feature
9	🕒 Account subscription design #818	Planning📅	azenMatt	★ Low	
10	🕒 Free and paid levels #816	Building🔨	JannesPeters	★ Low	need help

Data wrangling

Data wrangling deals with several functionalities:

1. Data exploration: In this process, the data is studied, analyzed and understood by visualizing representations of data.
2. Dealing with missing values: Most of the datasets having a vast amount of data contain missing values of NaN, they are needed to be taken care of by replacing them with mean, mode, the most frequent value of the column or simply by dropping the row having a NaN value.

Data wrangling

Data wrangling deals with several functionalities:

1. Data exploration
2. Dealing with missing values
3. Reshaping data: In this process, data is manipulated according to the requirements, where new data can be added or pre-existing data can be modified.
4. Filtering data: Some times datasets are comprised of unwanted rows or columns which are required to be removed or filtered

pandas and numpy

```
import pandas as pd
```

```
import numpy as np
```

pandas	numpy
When we have to work on Tabular data, we prefer the pandas module.	When we have to work on Numerical data, we prefer the NumPy module.
Pandas have a 2D table object called DataFrame.	Numpy is capable of providing multi-dimensional arrays.
The powerful tools of pandas are DataFrame and Series.	the powerful tool of NumPy is Arrays.
Pandas consume more memory.	Numpy is memory efficient.
Indexing of the Pandas series is very slow as compared to Numpy arrays.	Indexing of Numpy arrays is very fast.

pandas operation

read csv files into a pandas df: `pd.read_csv("link")`

Dataset

1. kaggle

<https://www.kaggle.com/datasets>

2. UCI Machine Learning Repository

<https://archive.ics.uci.edu/datasets>

3. Google Dataset Search

<https://datasetsearch.research.google.com/>

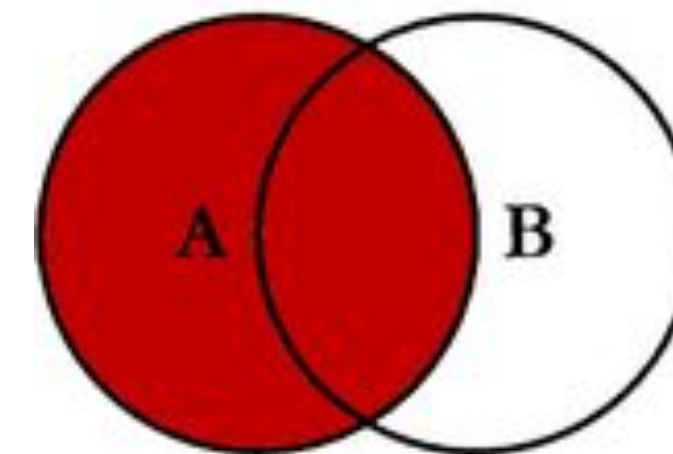
4. Github or other resources

Best practices when working with large data sets

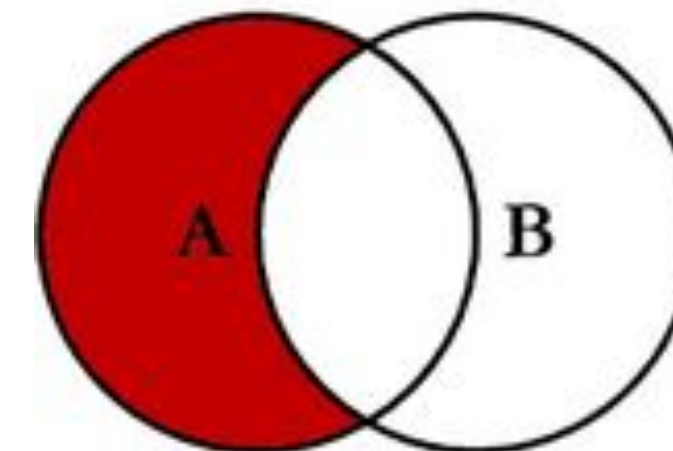
Be careful when joining, aggregating, sorting, or filtering your dataset.

Make sure to limit the dataset to only columns and rows needed.

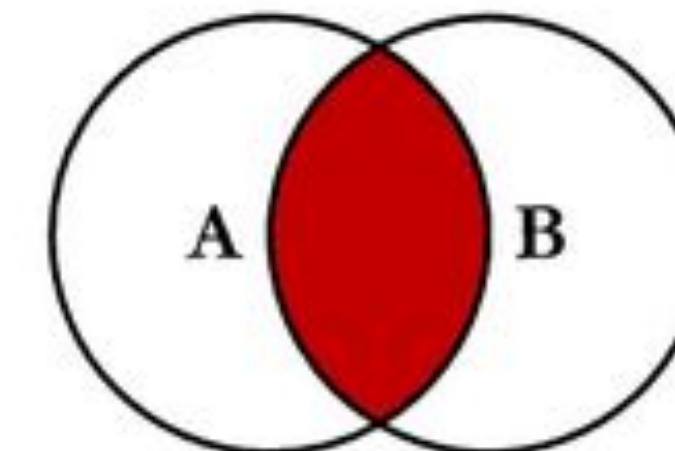
SQL JOINS



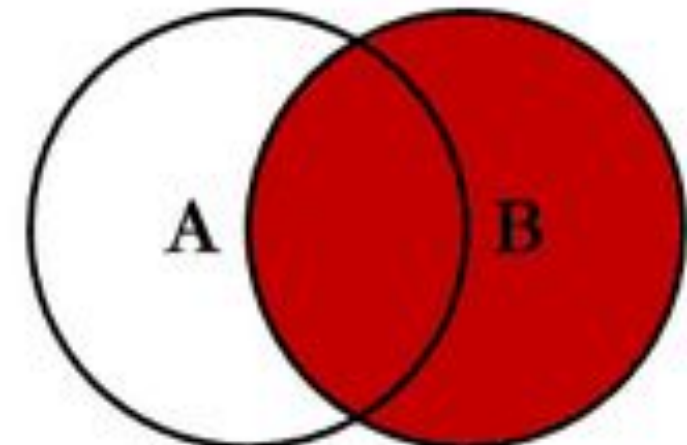
```
SELECT <select_list>  
FROM TableA A  
LEFT JOIN TableB B  
ON A.Key = B.Key
```



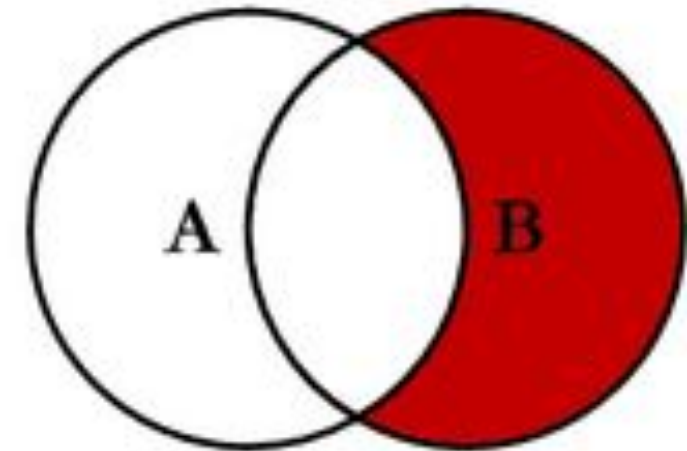
```
SELECT <select_list>  
FROM TableA A  
LEFT JOIN TableB B  
ON A.Key = B.Key  
WHERE B.Key IS NULL
```



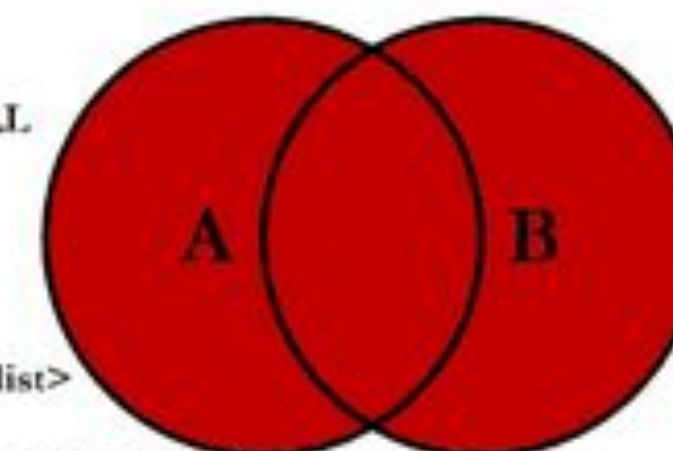
```
SELECT <select_list>  
FROM TableA A  
INNER JOIN TableB B  
ON A.Key = B.Key
```



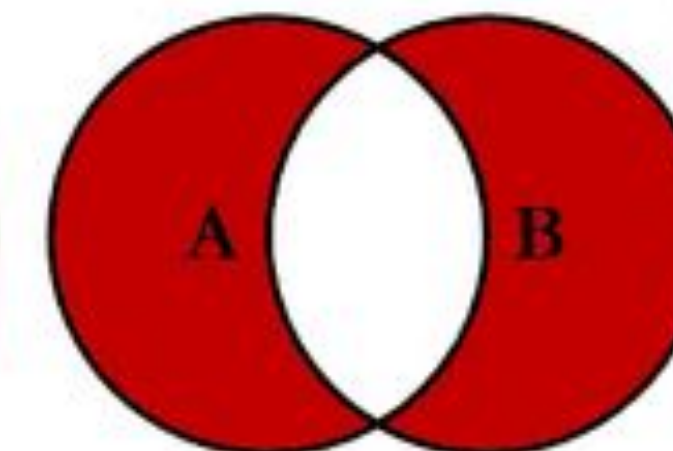
```
SELECT <select_list>  
FROM TableA A  
RIGHT JOIN TableB B  
ON A.Key = B.Key
```



```
SELECT <select_list>  
FROM TableA A  
RIGHT JOIN TableB B  
ON A.Key = B.Key  
WHERE A.Key IS NULL
```



```
SELECT <select_list>  
FROM TableA A  
FULL OUTER JOIN TableB B  
ON A.Key = B.Key
```



```
SELECT <select_list>  
FROM TableA A  
FULL OUTER JOIN TableB B  
ON A.Key = B.Key  
WHERE A.Key IS NULL  
OR B.Key IS NULL
```

Your time to ...

- Start looking for datasets for your final projects
- Work on D2